McCann model for rhomohedral graphene

April 23, 2026

1 Introduction

This report analyzes the behavior of ABC-stacked multilayer graphene using the McCann model. We specifically compare the cases for $N = 1$ (monolayer) and $N = 5$ layers, focusing on their energy bands, Berry curvature, and the convergence of the local Chern number contribution.

2 Model and Analytical Equations

The system is described by a 2x2 effective Hamiltonian in the vicinity of a K point (low energy):

$$H = \begin{pmatrix} \Delta & X(\mathbf{k}) \\ X^\dagger(\mathbf{k}) & -\Delta \end{pmatrix} \tag{1}$$

where 2Δ is the energy gap and $X(\mathbf{k})$ is the off-diagonal coupling term. In polar coordinates (k, ϕ), the term X for N layers is given by:

$$X(k, \phi) = \frac{(vke^{i\xi\phi})^N}{(-\gamma_1)^{N-1}} \tag{2}$$

where $v = \frac{\sqrt{3}}{2}\gamma_0$ in natural units ($a = 1$, $\hbar = 1$) and $\xi = \pm 1$ is the valley index.

2.1 Energy Bands

The eigenvalues of the Hamiltonian represent the energy bands:

$$\varepsilon(\mathbf{k}) = \pm E(\mathbf{k}) = \pm\sqrt{|X(\mathbf{k})|^2 + \Delta^2} \tag{3}$$

2.2 Eigenstates

The normalized eigenstates for the positive and negative bands are given by:

$$|\psi_+\rangle = \frac{1}{\mathcal{N}_+}\begin{pmatrix} E + \Delta \\ X^\dagger \end{pmatrix}, \quad |\psi_-\rangle = \frac{1}{\mathcal{N}_-}\begin{pmatrix} -X \\ E + \Delta \end{pmatrix} \tag{4}$$

where $\mathcal{N}_\pm$ are the normalization constants.

2.3 Berry Curvature

The Berry curvature Ω is calculated using the Kubo-like formula. For the results presented in this report, we specifically show the Berry curvature calculated for the positive energy band:

$$\Omega_n = i \sum_{m \neq n} \frac{\langle \psi_n | \nabla_\mathbf{k} H | \psi_m \rangle \times \langle \psi_m | \nabla_\mathbf{k} H | \psi_n \rangle}{(E_n - E_m)^2} \tag{5}$$

Here, the gradient is calculated in polar coordinates as:

$$\nabla_\mathbf{k} = \frac{\partial}{\partial k}\vec{e}_k + \frac{1}{k}\frac{\partial}{\partial \phi}\vec{e}_\phi \tag{6}$$

3 Numerical Results

3.1 Energy Bands Comparison

Figure 1 shows the energy bands for $N = 1$ and $N = 5$. For $N = 1$, the dispersion is linear at high k (Dirac-like), while for $N = 5$ it is much flatter near $k = 0$ due to the k^N dependence.

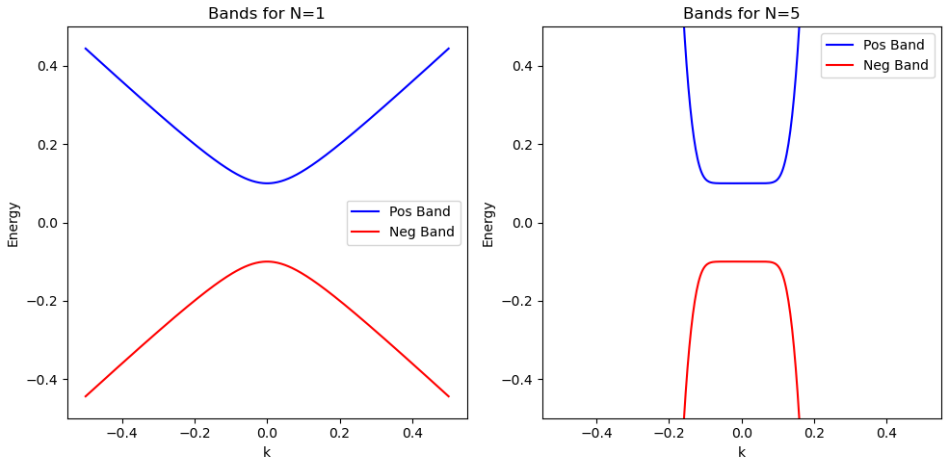

Figure 1: Energy bands for N=1 and N=5 layers.

3.2 Berry Curvature

The Berry curvature for $N = 1$ peaks sharply at $k = 0$. However, for $N = 5$, the curvature is suppressed at the center and peaks at finite k (Figure 2).

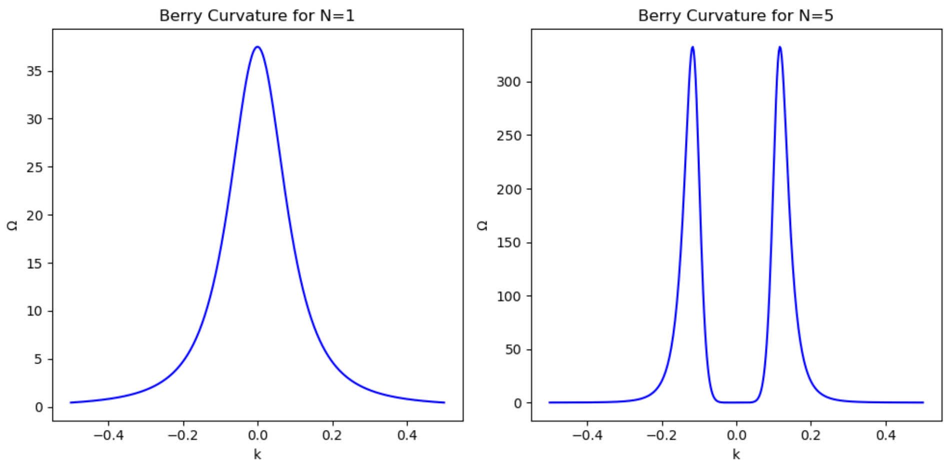

Figure 2: Berry curvature comparison (Positive Band).

4 Berry Curvature Integration

The local Chern number contribution is obtained by integrating the Berry curvature over a region in k-space. While the McCann model is isotropic and naturally expressed in polar coordinates, the numerical integration is performed over a uniform Cartesian meshgrid (k_x, k_y). This choice provides several numerical advantages:

- **Uniform Sampling Density:** In a Cartesian grid, every point represents an identical area element $\Delta A = \Delta k_x \Delta k_y$. In a polar grid, the area element $dA = k\,dk\,d\phi$ is proportional to k and vanishes as $k \to 0$. Since the Berry curvature is often highly concentrated near the Dirac point ($k = 0$ for monolayer graphene), the Cartesian grid ensures high sampling density where it is most needed without requiring a non-linear radial distribution.

- **Singularity Avoidance:** The Cartesian coordinate system avoids the coordinate singularity at the origin inherent to polar grids, which simplifies the handling of the peak in Ω at $k = 0$ for $N = 1$.

To ensure a consistent comparison between different layer numbers N, we define the integration domain using an energy cutoff E_{max} rather than a fixed momentum radius. The integral is evaluated numerically as:

$$C_{local}(E_{max}) = \frac{1}{2\pi} \iint_{E(\mathbf{k}) \leq E_{max}} \Omega_{pos}(\mathbf{k}) dk_x dk_y \approx \frac{\Delta A}{2\pi} \sum_{i \in \text{mask}} \Omega_{pos}(\mathbf{k}_i) \tag{7}$$

where the sum runs over all points k_i satisfying the energy condition $E(\mathbf{k}_i) \leq E_{max}$, and ΔA is the constant area per grid point.

Figure 3 shows that as the energy cutoff E_{max} increases, the integral converges to $N/2$.

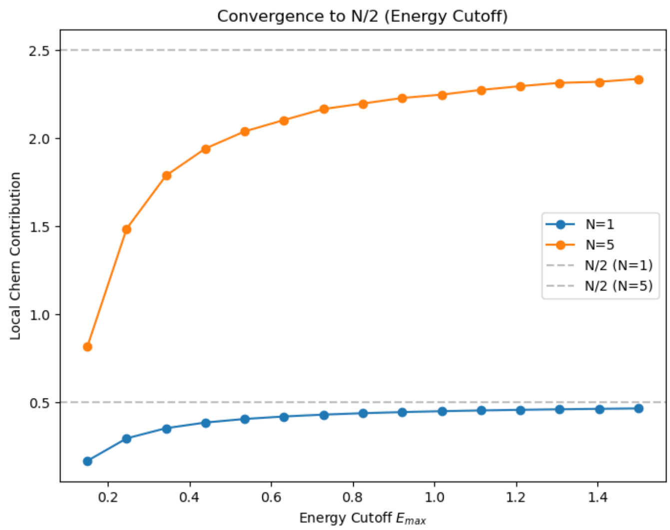

Figure 3: Convergence of the local Chern number contribution to $N/2$ as a function of the energy cutoff E_{max}.

5 Valley Comparison

The two K-valleys in the hexagonal lattice, indexed by $\xi = \pm 1$, are related by time-reversal symmetry. While the energy dispersion is identical for both valleys, the Berry curvature of a given band changes its sign:

$$\begin{aligned} \mathcal{T}: \; & \mathbf{k} \to -\mathbf{k} \\ & E(-\mathbf{k}) = E(\mathbf{k}) \\ & \Omega(-\mathbf{k}) = -\Omega(\mathbf{k}) \end{aligned} \tag{8}$$

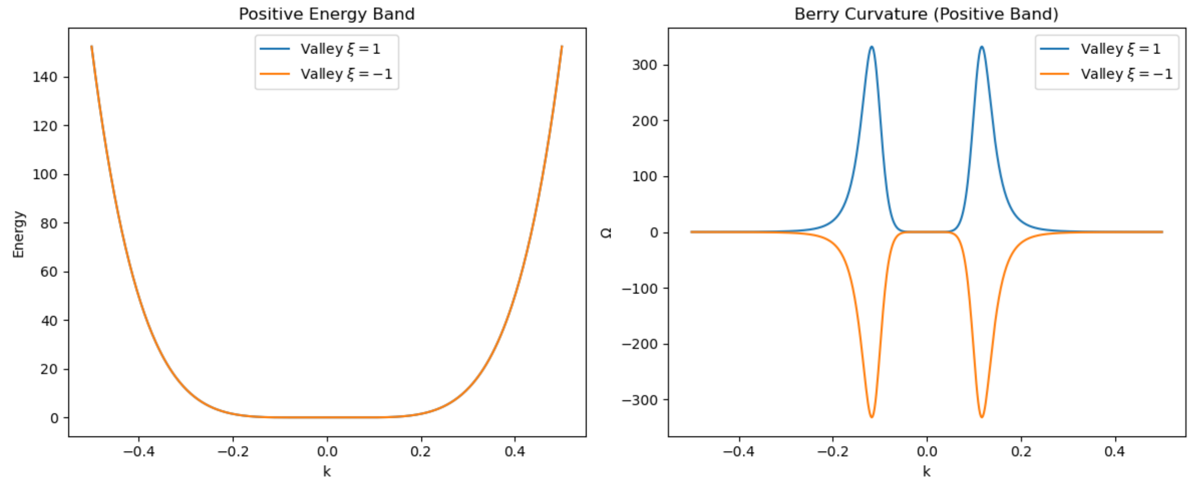

Figure 4: Comparison of the positive energy band (left) and Berry curvature (right) for valleys $\xi = 1$ and $\xi = -1$ ($N = 5$ layers).

Figure 4 compares the positive energy band and the corresponding Berry curvature for $N = 5$ layers, showing the sign inversion between the two K valleys.

Consequently, the local Chern number contributions for $N = 5$ have opposite signs. The integration of the Berry curvature over a circle of radius k=1 gives:

- Valley $\xi = 1$: $C_{local} = 2.499949$
- Valley $\xi = -1$: $C_{local} = -2.499949$

The total Chern number for the system, considering both valleys, would vanish in the presence of time-reversal symmetry ($C_{tot} = C_K + C_{K'} = 0$).

6 Anomalous Hall Effect with Spin

Recent additions to the project include the calculation of the Anomalous Hall Effect (AHE) considering both spin and valley degrees of freedom.

6.1 Magnetic AHE at $\mu = 0$

We calculate the total Hall conductivity σ_{xy} for the system at the Fermi level $\mu = 0$. We use $N = 5$ layers and a temperature $T = 20$ K. The gap parameters used are:

- Valley K ($\xi = 1$): $\Delta_\uparrow = 0.1$, $\Delta_\downarrow = 0.0$
- Valley K' ($\xi = -1$): $\Delta_\uparrow = 0.0$, $\Delta_\downarrow = -0.1$

The calculated total conductivity for the spin-valley case (without SOC) is:

$$\sigma_{xy}^{total}(\mu = 0) \approx -5.000\ e^2/h \tag{9}$$

which corresponds to the sum of the Chern contributions of the occupied bands ($N/2 = 2.5$ per valley per spin when applicable).

When including the Spin-Orbit Coupling ($\lambda = 10^{-3}\gamma_0$), the results at $\mu = 0$ remain consistent with the topological expectation:

$$\sigma_{xy}^{total,SOC}(\mu = 0) \approx -5.000\ e^2/h \tag{10}$$

as the SOC slightly shifts the gaps but the overall topology of the occupied manifold at $\mu = 0$ is preserved.

6.2 AHE vs. Chemical Potential

The dependence of the AHE on the chemical potential μ was scanned using the following parameters:

- Valley K: $\Delta_\uparrow = 0.1$, $\Delta_\downarrow = 0.0$
- Valley K': $\Delta_\uparrow = 0.0$, $\Delta_\downarrow = -0.2$

Figure 5 shows the results of the scan. It is important to note that the convergence to 0 far from $\mu = 0$ (as well as the convergence to $N/2$ when there is a plateau) is very slow and difficult to fully observe in numerical simulations due to the long tails of the Berry curvature.

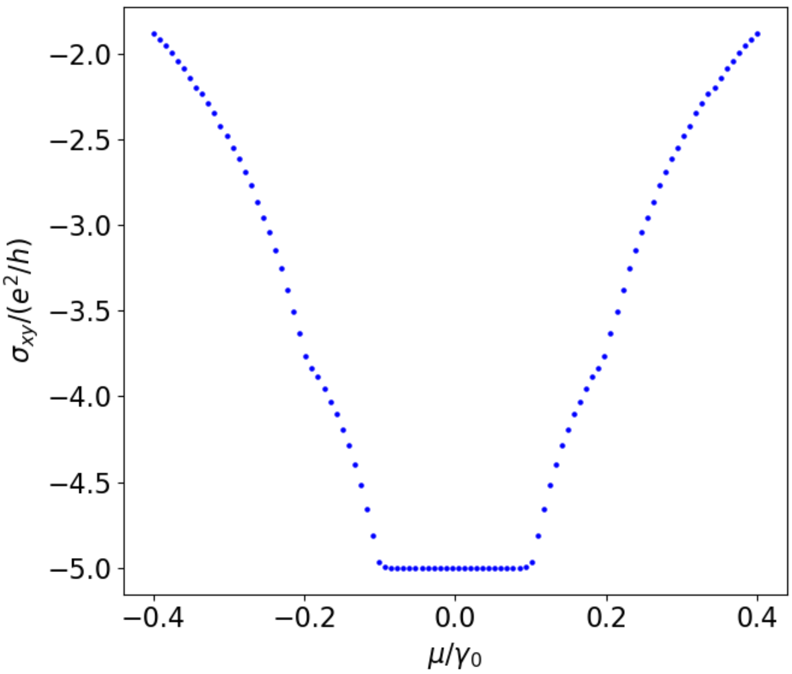

Figure 5: Total Hall conductivity σ_{xy} as a function of the chemical potential μ/γ_0 including spin and SOC.

7 Conclusion

The N-layer McCann system exhibits a topological Berry phase of $N\pi$, leading to a local Chern contribution of $N/2$ per valley in the gapped state. The distribution of the Berry curvature becomes increasingly non-trivial as N increases, with the peak shifting away from the Dirac point while the total topological charge remains conserved and proportional to N.